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Feedback Control of the Spatiotemporal Firing Patterns of Neural Microcircuits

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Abstract— **One of the fundamental objectives in systems neuroscience is to precisely control the spatiotemporal firing of cortical neurons to elicit a desired pattern of activity. In this work, we study the effects of intracortical micro-stimulation on the dynamics of a basal ganglia microcircuit model, and explore the feasibility of controlling the spatiotemporal firing patterns of the population in the presence of unobserved inputs. Results from the simulation study suggest that properly designed Multiple-Input-Multiple-Output (MIMO) feedback controller can force a subpopulation of output neurons to follow a prescribed spatiotemporal firing pattern despite the presence of unobserved inputs. The accuracy of the spike timing of the controlled neural firing with respect to the reference spike trains is in the order of tens of milliseconds. Even a simplified circuit model of Hammerstein-Wiener type can help in prescreening potential stimulation sites and analyzing the nominal stability of the closed-loop system.**

I. INTRODUCTION

Microstimulation has been an indispensable tool for identifying the functional organization of the brain and for establishing the causal links between neural activity and behavior. The ability of microstimulation to alter the firing patterns of neural circuits has already found a number of practical applications that go beyond basic scientific investigation, such as Deep Brain Stimulation (DBS) in the treatment of Parkinson's Disease (PD) [1], and cochlear implant for hearing restoration [2].

In many cases, it is highly desirable to adjust the instances and parameters of microstimulation in a closed loop setting using available measurements. The concept of feedback is not novel in biological neural systems. In fact, feedback exists almost everywhere in neural microcircuits (e.g. over expression of bidirectional links [3]) and is suggested to underlie many of its vital functions [4]. A number of computational [5], [6], [7] and biological studies [8], [9] have already demonstrated that feedback can be highly effective in regulating ensemble neural activities, particularly in the control of synchronized cell assemblies. Since the objective of the control is to desynchronize/regularize the whole population using a few macroelectrodes with broad spatial current/voltage spread, the feedback control design usually does not take into consideration the network topology and the complex dynamic interactions among neural elements. As a result, they have limited generalizability to the cases in which precise spatiotemporal control is desired.

With the rapid advance of microelectrode arrays [10] and combined optical stimulation and electrophysiological recording techniques [11], [12], we now can simultaneously stimulate and record the activities of hundreds of neurons at a spatiotemporal resolution that is almost impossible to imagine decades ago. This opens up new opportunities for precise spatiotemporal control of neural activity at both single-cell and population levels. In an intricately connected neural circuit, achieving a high spatial (single cell) and temporal (millisecond) resolution by selectively stimulating certain neural elements, however, has shown to be a nontrivial task, especially in the presence of many unobserved inputs and disturbances. To what extent the spatiotemporal firing patterns at a single cell level can be effectively controlled and how this could be related to the synaptic organization of the circuit, will be the focus of this study.

II. MODELING NEURAL MICROCIRCUITS

The synaptic connectivity and the dynamics of participating neural elements, shape the dynamics of a neural microcircuit in response to stimulations (natural or artificial), and decide how information is processed and transmitted across the network. To create a realistic neural microcircuit, there are two major pieces of information we need to consider: the synaptic connectivity and the dynamics of the participating neural elements including cells and synapses.

Our knowledge about the synaptic organization is still incomplete [13], mainly due to the experimental difficulties involved in identifying those connections and their relative strength at such a detailed micro level for a large population of neurons. Most of the computational models of microciruits today have to rely on the statistical information about the synaptic projections among distinct neural populations obtained from anatomical studies [14], [15].

Compared with the synaptic organization, the electrophysiological properties of various types of excitable membranes and synapses are rather well documented in terms of in vitro observations and experimentally fitted computational models, especially after the pioneering work by Hodgkin and Huxley[16], [17]. The building blocks are the kinetic models of ion channels. At a microscopic level, a single ion channel undergoes rapid stochastic transitions between conducting and nonconducting states, and the stochastic transitions between different states are well fitted by Markov models

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with time dependent transition probability that is potentially dependent on membrane potential, calcium concentration, transmitter or second-messenger [18]. For example, the simplest Markov model of a single ion channel is

$$C \underset{\beta(X)}{\overset{\gamma(X)}{\rightleftarrows}} O$$

where X is the extrinsic variable to the channel such as voltage and neurotransmitter concentration, $\gamma(X)$ and $\beta(X)$ are the extrinsic variable-dependent transition rates between the closed and the open state. The temporal evolution of the probability $P^o(t)$ of the open state follows the kinetic equation

$$\frac{dP^o}{dt} = (1 - P^o) \cdot \gamma(X) - P^o \cdot \beta(X)$$

The macroscopic behavior of the ionic currents can then be modeled as

$$i = g \cdot P^o \cdot (V - E) \tag{1}$$

where g is the maximum conductance when all the corresponding ion channels are open, V is the membrane potential and E is the corresponding reversal potential.

By choosing appropriate Markov model structures and the dependence of the transition rates on extrinsic variables, one can model a wide range of dynamic behaviors exhibited by ion channels. Different types of cells and synapses can then be constructed by combining an appropriate set of ion channels to reproduce experimentally observed electrophysiological phenomena. In summary, for a microcircuit consisting of N neurons, we have

$$C_i \dot{V}_i = -\sum_k I_k^{ion} - \sum_{j \neq i} I_{i,j}^{syn} + I_i^{inj}, i = 1, 2, \ldots, N \tag{2}$$

where C_i and V_i are the membrane capacitance and potential of neuron i, respectively. I_k^{ion} is the ionic current, $I_{i,j}^{syn}$ is synaptic current from neuron j to neuron i, and I_i^{inj} is the intracellularly injected current at neuron i. The ionic current and the synaptic current are both of the general form (1) in which P^o is potentially time and extrinsic-variable dependent.

The effects of microstimulation are typically modeled as additional ionic currents. For example, the stimulation induced ionic current can be modeled as

$$i^{sti} = g \cdot \Delta P^{sti} \cdot (V - E) \tag{3}$$

where $0 \leq \Delta P^{sti} \leq 1$ is potentially time, voltage and stimulation parameter dependent.

Since our focus is to study the network effects of microstimulation, we will assume the effect of micorstimulation is manifested through a prototyped current pulse. The control variable is thus the stimulation frequency. The control problem considered in this paper is to track the spatiotemporal firing of a subset of neurons by stimulating a number of potential stimulation sites as illustrated by an example in Fig. 1 and Fig. 2.

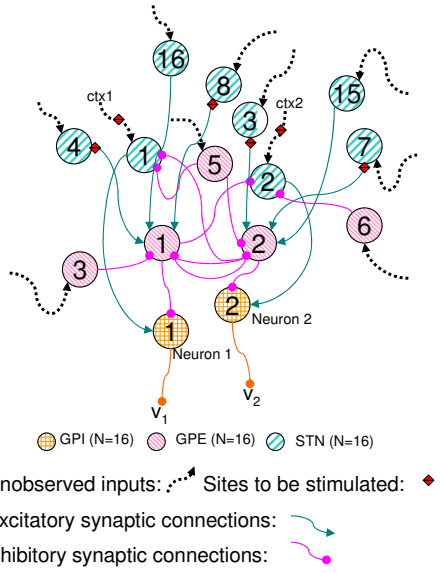


Fig. 1. The connectivity patterns among neurons in a portion of the simulated BG microcircuit

III. METHODS

A. The Benefit of Feedback

A Matlab/Simulink model is implemented based on parameters provided in [14]. The model consists of three populations of neurons: STN (Subthalamic Nucleus), GPe (Globus Pallidus externalis), and GPi (Globus Pallidus internalis). Each population consists of 16 single-compartment Hodgkin-Huxley neurons. Constant currents are injected into each neuron to make them fire spontaneously at around 30, 60, and 80 spikes per second for STN, GPe and GPi neurons, respectively. Each STN neuron provides excitatory inputs to a subpopulation of GPe and GPi neurons. In addition to the self inhibitory connections within the GPe neuron population, they also inhibit neurons in the STN and GPi population. Figure 1 shows a portion of the model that has direct or 1st-level-indirect synaptic projections to GPi neuron 1 and GPi neuron 2. We choose to control the firing patterns of GPi neurons because the activity of GPi neurons constitutes a major output of the Basal Ganglia (BG) system [19]. Under this structure, the rhythmic firing caused by circuit disorder in basal ganglia system is manifested through the rhythmic firing of GPi neurons, which is commonly considered to be the cause of certain symptoms of PD, such as tremor. One deviation from the model in [14] is that we let each STN neuron accept an independently generated spike train to mimic the excitatory cortical input.

Before going further into the issues about stimulation sites selection and closed loop stability, we first demonstrate the effectiveness of output feedback in tracking the firing patterns of the output neurons. Figure 2 shows the high level control architecture used in the simulation to control the firing patterns of both output GPi neurons. The spike trains are first extracted from the membrane potentials and passed through a Finite Impulse Response (FIR) filter. The

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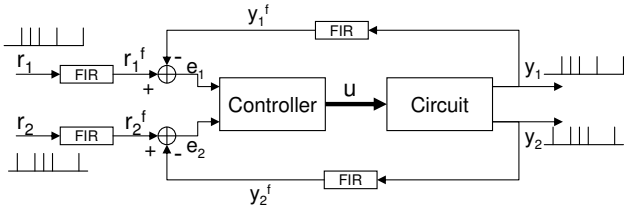


Fig. 2. The structure of the controller used to control the firing patterns of two output GPi neurons

FIR filter in our simulation is simply a sliding window that computes the firing rate of the neurons,

$$r(t) = \frac{1}{K \cdot \Delta T} \sum_{k=0}^{K-1} y(t-k) \tag{4}$$

where $y(t)$ is the spike train (equal to one if there is a spike at time t, zero otherwise), K is the size of the window and ΔT is the sample time. $K \cdot \Delta T$ is set to 200 milliseconds throughout the paper. The differences (e_1 and e_2, respectively) between the filtered reference spike trains (r_1^f and r_2^f) and the filtered output spike trains (y_1^f and y_2^f) are then used to generate the stimulation pulses that are delivered to different sites as marked in Fig. 1 using $\diamondsuit$.

The Multiple-Input-Multiple-Output (MIMO) controller is of a simple PID type. To simplify the parameter tuning, the controller has an internally decoupled structure that uses only e_1 to control the firing patterns of GPi neuron 1 by stimulating sites that influence STN neuron 1 and GPe neuron 1. And similarly uses only e_2 to control the firing patterns of GPi neuron 2. Figure 3 shows the tracking performance of the MIMO controller for controlling the spatiotemporal firing patterns of the two GPi neurons. The Variance Accounted For (VAF) ($100\%(1-\frac{var(r^f-y^f)}{var(r^f)})$) between the filtered reference spike trains (r_1^f and r_2^f) and the filtered actual spike trains (y_1^f and y_2^f) is 87% and 85% for GPi neuron 1 and 2, respectively. Even though not every spike in the reference spike train is reproduced, the controller certainly forced the output neurons to fire at a similar pattern and the accuracy of the spike timing is roughly in the order of tens of milliseconds on average. Further cross correlation analysis (two lower plots in Fig. 3) reveals that there are systematic delays between the references and the outputs, the time lags are around 15 ms and 12 ms for GPi neuron 1 and 2, respectively. After the time lags are compensated by delaying the reference spike train with respect to the actual evoked spike train, the VAF becomes 93% for GPi neuron 1 and 90% for GPi neuron 2.

The reference spike trains are independently generated through a modified Poisson process with periodically reinforced silence period, and yet the simple MIMO feedback controller is able to force the output neurons to fire at a delayed but similar pattern. Since the effects of stimulation are modeled as additional prototyped current pulse injection, the unobserved inputs can still pass through from presynaptic neurons to postsynaptic neurons without much interruption unless the injection frequency is very high. Even in this small

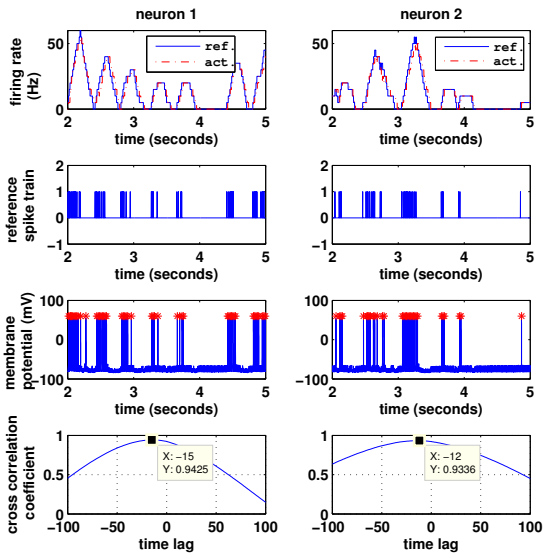


Fig. 3. Tracking the spatiotemporal firing patterns of two output neurons

portion of the circuit shown in Fig. 1, there are large number of unobserved inputs that either directly or indirectly affect the firing patterns of the two output GPi neurons.

B. A Simplified Model

A comprehensive analysis of the dynamics of the neural circuits is considerably more complicated than the corresponding analysis of their structure. Even with the perfect knowledge of the synaptic organization, the dynamics of the network can not be completely determined due to the complex nonlinear dynamic interactions among the participating neural elements. And the direct stability analysis on the detailed circuit model as in (2) with spiking neurons can be extremely challenging. For the purpose of advanced control design, we need a substantially simplified model that captures the essential dynamics of the circuit and allows better analytical tractability.

Recent work suggests that subsequent single neuron spiking can be predicted with substantial accuracy using a generalized linear model (GLM) based on the intrinsic and ensemble spiking history [20]. Although this point process-GLM framework [21] offers advantage over conventional linear or nonlinear regression methods for directly modeling the point process time series without smoothing the original discrete event observations, its structure bears unmistakable similarity with Hammerstein-Wiener model [22] in the sense that both models use a linear model structure coupled with static input/output nonlinearity. Models of similar structures have been used to model cell and population firing activities [23], [24], [25]. Since it is much more convenient to deal with the control design problem of continuous-valued systems, one of plausible choices is to use the following simplified model to identify the neural circuits.

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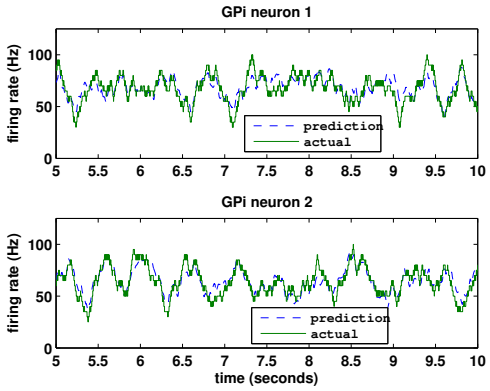


Fig. 4. Comparison of the predicted and actual firing rate using the identified circuit model. The VAF is 64% and 62% for GPi neuron 1 and GPi neuron 2, respectively.

$$\begin{aligned} x_i(t) &= \sum_{j\neq i} \frac{B^r_{i,j}(q^{-1})}{F^r_{i,j}(q^{-1})} r_j(t) + \sum_j \frac{B^u_{i,j}(q^{-1})}{F^u_{i,j}(q^{-1})} u_j(t) + e_i(t) \\ r_i(t) &= h(x_i(t);\theta_i) \end{aligned} \quad (5)$$

for $i = 1, \ldots, N$ where N is the number of neurons. r_i is the firing rate of the neuron i over a sliding window of length $T = K \cdot \Delta T$ as in (4), $u_j(t)$ is the stimulation input j, q^{-1} is the unit delay operator, $x_i(t)$ is an intermediate variable, the term $\frac{B^r_{i,j}(q^{-1})}{F^r_{i,j}(q^{-1})}$ models the dynamic influences of the firing rate of neuron j on the firing rate of neuron i, and the term $\frac{B^u_{i,j}(q^{-1})}{F^u_{i,j}(q^{-1})}$ captures the dynamic interactions between input u_j and the firing rate of neuron i. $h(x_i(t);\theta_i)$ is the static output nonlinearity parameterized by θ_i, which, e.g., can be used to take care of the background level of spiking rate and the nonlinearity in the spiking characteristic curve.

For the purpose of simplicity, only a small portion of the circuit, including three pairs of STN (1 and 2), GPe (1 and 2) and GPi (1 and 2) neurons, is identified using the firing rates computed from the spike trains based on (4). Figure 4 shows the predicted firing rates of GPi neuron 1 and GPi neuron 2 under the same the inputs (in terms of firing rates) from the rest of the circuit. The nonlinearity used in the identified model is a first order polynomial of the form $k_i x_i(t) + b_i$, and higher order polynomials or other types of nonlinearities such as sigmoidal functions do not significantly improve the modeling performance.

It is generally not advisable to isolate a portion of the circuit to analyze, since there might be strong feedback loops between the isolated portion and the rest of the network. It is justifiable, however, when the selected portion of the circuit has strong/reciprocal internal connections and weak/unidirectional external connections, which is likely to be the case in actual biological neural circuits [3], [26] and the dynamical properties of the network motifs might play an important role in deciding the circuit dynamics [27]. In the network structure proposed in [14], the synaptic connection pattern among the six neurons is repeated throughout the rest of network. Therefore, analyzing the dynamic properties of this network motif can offer some insights into the design of the MIMO controller.

TABLE I

THE RELATIVE GAIN ARRAY OF THE TRANSFER FUNCTIONS

	From					
To	ctx2	ctx1	stn4	stn8	stn3	stn7
GPi1	0	0.0008	0.3029	0.6872	0.0032	0.0089
GPi2	0.0006	0	0.0037	0.0084	0.2600	0.7303

C. Stimulation Site Selection and Nominal Stability

Facing a multitude of potential stimulation sites in the circuits, it is naturally of great interest to select the few that are most effective in controlling the outputs. There are a number of input-output selection methods which are commonly used in the control design for MIMO systems [28]. The frequency-domain non-square relative gain array (RGA) is often employed to eliminate the inputs whose column sum is much smaller than one when the number of inputs is greater than the number of outputs [29, page 90]. Table I shows the RGA matrix for the identified circuit model. In this case, the values in the RGA array do not change significantly with respect to frequency. As it is suggested by the criteria, the first two inputs can be removed. With the controller parameter unchanged in the detailed simulation as in Sec. III.A, stimulating the four remaining sites achieves VAF of 86% and 86% for GPi neuron 1 and neuron 2, respectively. After a careful investigation of the detailed circuit model, we found that the synaptic connections along the shortest paths from stn4 and stn8 to gpi1, or from stn3 and stn7 to gpi2 are much stronger.

Although the non-square RGA is an efficient prescreening tool, other factors still need to be considered regarding the controllability, such as input saturation. For example, if we only pair up stn8 with gpi1 and stn7 with gpi2, the tracking performance can significantly degrade mainly due to actuator saturation, since we assume the upper limit for the stimulation is $200Hz$. The information about the connectivity of the neural circuitry can also help to choose the most effective stimulation sites. Naturally, stronger connections with lower synaptic latency between the stimulation sites and the output neurons are often preferred if faster and more precise tracking is desired. Indeed, this type of graph based input-output selection methods do exist [28, Sec. 3.1]. There are techniques that can help identify the functional connectivity of the neural circuitry using available observations. One of the promising approaches is to use Dynamic Bayesian Networks (DBN) to infer the effective connectivity from observed spike trains collected using high-density microelectrode arrays [30].

Suppose we pair up stn4 and stn8 with gpi1, and stn3 and stn7 with gpi2 using the following decoupled control

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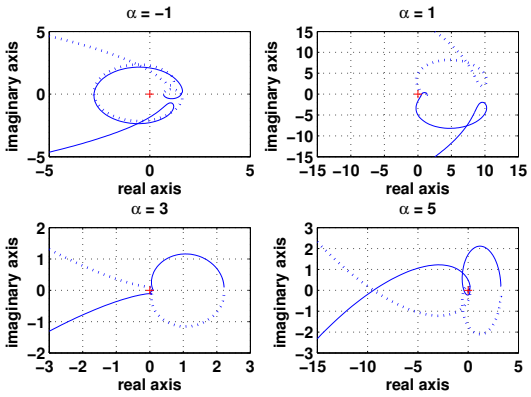


Fig. 5. Nyquist plots of $det(I + \alpha G^C(s)K^C(s))$ for $\alpha = -1, 1, 3, 5$

structure $K(q^{-1})$

$$\begin{pmatrix} K_0(q^{-1}) & 0 \\ K_0(q^{-1}) & 0 \\ 0 & K_0(q^{-1}) \\ 0 & K_0(q^{-1}) \end{pmatrix}$$

where $K_0(q^{-1}) = P_0^P + P_0^I \frac{T_s q^{-1}}{1-q^{-1}} + P_0^D(1 - q^{-1})$. Given $P_0^P = -1$, $P_0^I = -2$, and $P_0^D = -1$, the Nyquist plots of $det(I + \alpha G^C(s)K^C(s))$ for different values of α are given in Fig. 5, where $G^C(s)$ and $K^C(s)$ are the continuous time transfer functions converted from their discrete time counterparts $G(q^{-1})$ and $K(q^{-1})$ using bilinear approximation[1]. Based on the generalized Nyquist criterion for MIMO system [29, page 152], the closed-loop system is stable only for a small range of α from above 0 to around 3.5. Setting $\alpha = -1$ or $\alpha = 5$ both result in nonzero clockwise encirclements of the origin, which indicates that the closed-loop system is unstable because the loop transfer function $G^C(s)K^C(s)$ does not have poles in the open right half complex plane. Applying the multivariable circle criterion [31, page 265] also indicates that the closed loop system is absolutely stable for any memoryless, potentially time varying, nonlinear functions $\alpha(t, \cdot) \in (0.01, 3.5)$.

This observation is qualitatively consistent with the detailed simulation results, in which negative α dramatically degrades the tracking performance. Large positive α beyond 3.5, however, does not result in immediate degradation of the tracking performance as shown in Fig. 6. Nevertheless, further increasing α beyond 1 does not lead to notable improvement in the tracking performance, and the analytical results obtained from the simplified model still cover a useful range of α from which we can choose without potentially compromising the tracking performance. Several factors might contribute to this difference between the stability analysis based on the simplified model and the actual simulation, such as unmodeled nonlinear dynamics, actuator saturation and unmodeled feedback loops between the isolated network motif and the rest of the circuit.

[1]Mainly for the convenience of interpretation and easy accessibility of related literature on the stability analysis of MIMO systems in continuous time domain

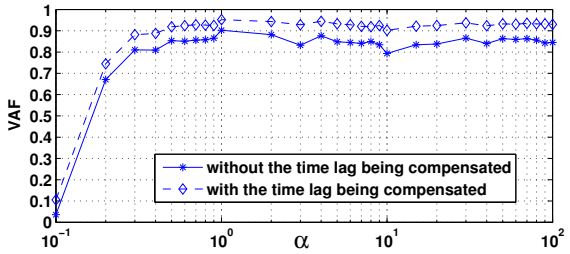


Fig. 6. The average VAFs between GPi neuron 1 and GPi neuron 2, with and without the time lags being compensated, are calculated based on 30 seconds simulation

IV. CONCLUSIONS AND FUTURE WORKS

The study presented in this paper suggests that properly designed MIMO feedback controller can force a subpopulation of observed output neurons to follow a prescribed spatiotemporal firing pattern despite the presence of unobserved inputs. The accuracy of the spike timing of the controlled neural firing with respect to the reference spike trains is in the order of tens of milliseconds. Even though it remains questionable whether a simplified model such as a Hammerstein-Wiener model can be effective in modeling large scale neural circuit, it certainly helps in prescreening potential stimulation sites and analyzing the nominal stability of the closed-loop system once a controller structure is given.

Although one can design a controller through trial and error to achieve certain performance specification without the knowledge of a simplified model, it can quickly becomes unmanageable as the number of stimulation sites and output neurons becomes large. A more systematic control design approach, such as model based design, which takes into consideration the circuit topology and the dynamic interactions, might be beneficial to achieve more satisfactory tracking performance. More advanced modeling techniques, especially the ones that take into consideration multiple time scales and essential nonlinearities [31], are desperately needed to achieve a balance between the modeling accuracy and the mathematical tractability.

There are several factors that can potentially limit how much we can do to control the spatiotemporal firing patterns of the neural circuits. Firstly, the spatial extent of the stimulation is heterogeneous and the pattern of activation is usually sparse. For example, in the case of extracellular electrical stimulation, the local composition and projections of neural elements, such as dendrites, soma and axons, can affect the spatial pattern of activation in electrical stimulation [32]. It is generally difficult to excite individual neurons. Secondly, the synaptic latency can be significant over long and indirect synaptic connections, which can impose a serious limitation on achievable control performance. Thirdly, it is unavoidable that the stimulation can introduce changes in the firing patterns of intermediate neurons, which might cause unwanted side effects.

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Some factors can play to our advantage. A study of rat visual cortex [3] suggests that the network can be viewed as "a skeleton of stronger connections in a sea of weaker ones". The strongest 17% of the connections contribute half of the total synaptic strength, which suggests that the strong ones may play more central roles in deciding the information processing and transmission of the neural circuit. The correlations in the strength of incoming and outgoing connections also suggest the presence of neurons with particularly strong connections, or "network hubs" [3]. The existence of such hub neurons has been recently demonstrated in [33], and the entire network dynamics can be influenced by perturbation of a single hub neuron. Therefore, identification of the exact connectivity of this skeleton of strongly connected neurons, and the population of the hub neurons is likely to greatly simplify the resulting microcircuit models and facilitate the control of the dynamics of the microcircuit. This skeleton of strong and reliable synaptic connections may have significant impact on the network dynamics and may make the firing patterns of the involved neurons highly reproducible for a given stimulation. If these strong synaptic connection patterns or network motifs can be reliably identified, even with less reliable identification of the weaker ones, we may still be able to capture the essential dynamic features of the microcircuit. By discarding some weaker connections, we may also significantly reduce the connectivity complexity of the resulting microcircuit.

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